

Marcus Chen

Full-Stack Developer · Distributed Systems · Open Source

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SUMMARY

Full-stack engineer with 8+ years building resilient web platforms, distributed services, and developer tooling. Previously led infrastructure teams at scale-stage startups; currently maintain several open-source projects (2.4k+ GitHub stars across them). Comfortable across the stack — from edge-deployed Rust services to pixel-level React work — and biased toward correct, boring, well-tested systems.

EXPERIENCE

Senior Staff Engineer · Helix Labs (independent)

2023 — present

Remote

- Designed and built **Helix**, an embedded ACID-compliant realtime database with row-level subscriptions; now powers several small SaaS products in production.
- Shipped **Tideflow**, a typed workflow orchestrator with durable execution and content-addressed replay; adopted by 4 teams handling ~12M job-runs/month.
- Contributed upstream to *tokio*, *sqlx*, and *astro*; merged 14 PRs across the Rust ecosystem.

Senior Engineer, Platform · Northwave (Series B, observability)

2020 — 2023

Brooklyn, NY

- Architected the edge analytics pipeline (later open-sourced as **Sparrow Edge**) streaming events from 40+ POPs into a ClickHouse cluster with sub-second freshness.
- Led a 4-engineer team rewriting the realtime ingest layer from Node to Rust; p99 latency dropped from 380ms to 42ms and infra cost fell by 61%.
- Owned the customer-facing dashboard (React/Next.js, GraphQL); raised Lighthouse perf from 64 → 96 and shipped WCAG 2.1 AA conformance.

Software Engineer · Loom Labs (Seed → Series A, devtools)

2017 — 2020

San Francisco, CA

- Joined as 6th engineer; built core ingestion and billing systems in Go + PostgreSQL, scaling from 200 → 28,000 daily active workspaces.
- Designed **Drift Migrate**, the open-source online-safe schema migrator now used internally and externally; replaces hand-rolled migration code for ~60 teams.
- Mentored 5 engineers and led the platform team's on-call and incident review process for two years.

OPEN SOURCE

Helix	Realtime embedded database — Rust, tokio, WAL	github.com/marcuschen/helix
Tideflow	Typed durable workflow orchestrator — Go, PostgreSQL, gRPC	github.com/marcuschen/tideflow
Lumen UI	Headless accessible React primitives — TypeScript, ~1.4k weekly d/I	github.com/marcuschen/lumen-ui
Quill CMS	Flat-file, git-backed CMS — Node.js, Astro, Yjs	github.com/marcuschen/quill

Sparrow Edge	Edge analytics pipeline — Rust, ClickHouse, Cloudflare	github.com/marcuschen/sparrow
Drift Migrate	Zero-downtime PostgreSQL schema migrator — Go	github.com/marcuschen/drift

SKILLS

Languages	TypeScript, JavaScript, Rust, Go, Python, SQL
Frontend	React, Next.js, Astro, CSS / Tailwind, Web Performance, WCAG / a11y
Backend & Data	Node.js, Deno, PostgreSQL, Redis / Streams, gRPC / Protobuf, GraphQL, Kafka, ClickHouse
Infra & Ops	Docker, Kubernetes, Terraform, AWS, GCP, Cloudflare, OpenTelemetry

SELECTED WRITING

- [2026.04](#) We rewrote our hot path in Rust — what we got, and what we didn't
- [2026.02](#) A pragmatic guide to durable workflows without the framework
- [2025.11](#) PostgreSQL row-level locking: when SELECT FOR UPDATE bites back

EDUCATION

B.S. Computer Science · University of Washington · 2013 — 2017

Focus: distributed systems, programming languages. Capstone: a verified key-value store with crash-consistent recovery.